Technology Study Memo

Team Members:

John “Yianni” Bray

Logan Hammer

Camila Martinez

Dylan Moga

Christian Warson

Topic: Propulsion and recovery overview. Analyzing different rocket motors and parachute systems.

Due Date: September 19th, 2025

Senior Design 1: EAS 3810

Tuesday, Thursday 3pm - 4:15pm

Group: 2



UNIVERSITY OF CENTRAL FLORIDA

Introduction

The University of Central Florida Controlled Recovery And Safe Handling (CRASH) Challenge is a challenge for students in senior design to model, manufacture, and launch a rocket with the purpose of transporting a payload and safely returning to the ground. Looking at launch and recovery of the rocket, sufficient propulsion as well as the use of a correct parachute is of utmost importance for reaching the desired altitude, slowing down the rocket prior to landing, and ensuring the rocket does not drift too far from the launch site. This memo will be focused on background info for the propulsion and recovery systems of a model rocket that will assist us in meeting the project requirements for our rocket. This memo will look at prior work in model rocketry as well as data on successful launches.

Propulsion

Most model rocket propulsion is typically achieved through the use of solid rocket motors which our team will be using as well. The motors most commonly used in model rocketry are produced by Estes, Aerotech, and Cesaroni. Aerotech and Cesaroni both offer reloadable motors, which may be beneficial in the future when our team has to take budgeting into account, but for now, that will not be considered.

Rocket motors are classified according to their total impulse, with low-power motors typically being A-D, mid-power motors being E-G, and high-power motors being H and above. As determined by the project requirements, the impulse limit of the rocket engine we select is a class H. However, due to the altitude requirements of the challenge restricting the maximum apogee of our rocket to 1500 feet, we will likely use a motor in the E-G range. On top of ensuring we do not exceed the maximum altitude of 1500 feet, a motor in this range will also

ensure that we meet the minimum altitude requirement of 1000 feet. Additionally, the challenge dictates that our rocket may not land further than 800 feet away from the launch site, so a mid-power rocket motor that propels our rocket to approximately 1200 feet will drift less than a higher powered motor that may propel our rocket to 1500 feet. Due to the rocket's need to carry a payload to the required altitude, our team is most likely to select either an F or G motor, as increased weight from the payload will necessitate the need for higher lift provided by the motor, to keep our rocket's thrust to weight ratio as close to 5:1 as possible, as recommended by the Tripoli Rocketry Association. Depending on how large our team can make the payload on our rocket, we may use an H motor, though this will require an L1 certification for purchase.



Fig 1. Motor size comparison

Motors come in multiple sizes, with the motors in the E-H range coming in the following diameters: 24mm, 29mm, 38mm. A larger diameter motor will require a larger diameter rocket, which will also allow us to fit a larger payload inside the rocket. Figure 1 shows five different motor sizes next to each other. It is important to note that as motor power increases, the body of the motor also increases in length. This means that a more powerful motor will result in less

cargo space within the body of the rocket. A way to solve this issue would be to use a larger diameter motor, which will allow for a shorter rocket motor of the same power.

MOTOR PERFORMANCE DATA

MOTOR TYPE	PROPELLANT WEIGHT		TOTAL IMPULSE (TYP)		AVERAGE THRUST	
	oz.	gms	lb.-sec	N-sec.	lbs.	N
F26FJ	1.52	43.1	14.6	65	5.8	26
F25W	1.37	38.8	16.4	73	5.6	25
F30FJ	1.12	31.8	10.8	48	6.7	30
F32T	0.91	25.8	13.3	59	7.2	32
F50T	1.19	33.6	15.7	70	11.2	50

TYPICAL TIME-THRUST CURVES

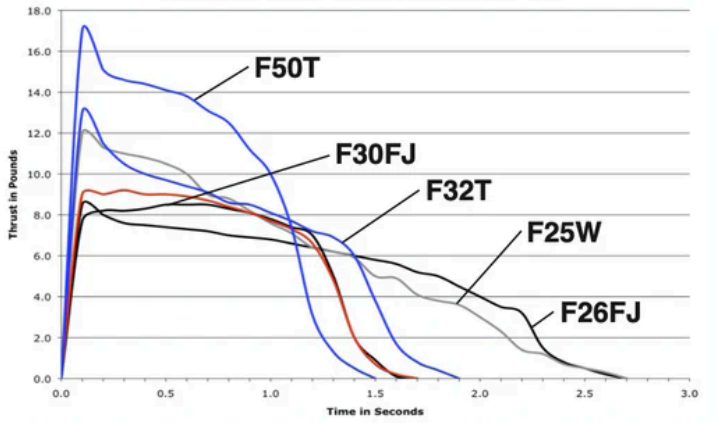


Fig 2. Motor Performance Data and Thrust curves

Another aspect of the motor that our team wants to look into is the burn time. A rocket motor with a short burn time will have more instantaneous thrust, which will lead to faster acceleration, but will also cause increased stresses on the payload and airframe which increases the chance for failure on launch. A rocket with a long burn time will accelerate slower and smoother, leading to reduced structural stress and a lower likelihood of airframe failure or payload damage. Figure 2 lists some motor specifications and graphs the thrust versus time of each motor. In the “Total Impulse” section in the table we can see that the total impulse of each motor is relatively close. However in the graph we can see that the F50T uses the majority of its power in the first 0.2 seconds of flight. This thrust curve would be suboptimal for our rocket as it

could possibly lead to damage to our payload. A thrust curve similar to the one of the F26FJ would be significantly more desirable as the acceleration over time would reduce the chances of damage to the payload.

From all the data collected, it seems that an AeroTech H195NT-14A 29mm motor would provide the most thrust and lead to the ability to carry the greatest amount of eggs in the cargo bay of our rocket, while also pushing the rocket to the desired altitude.

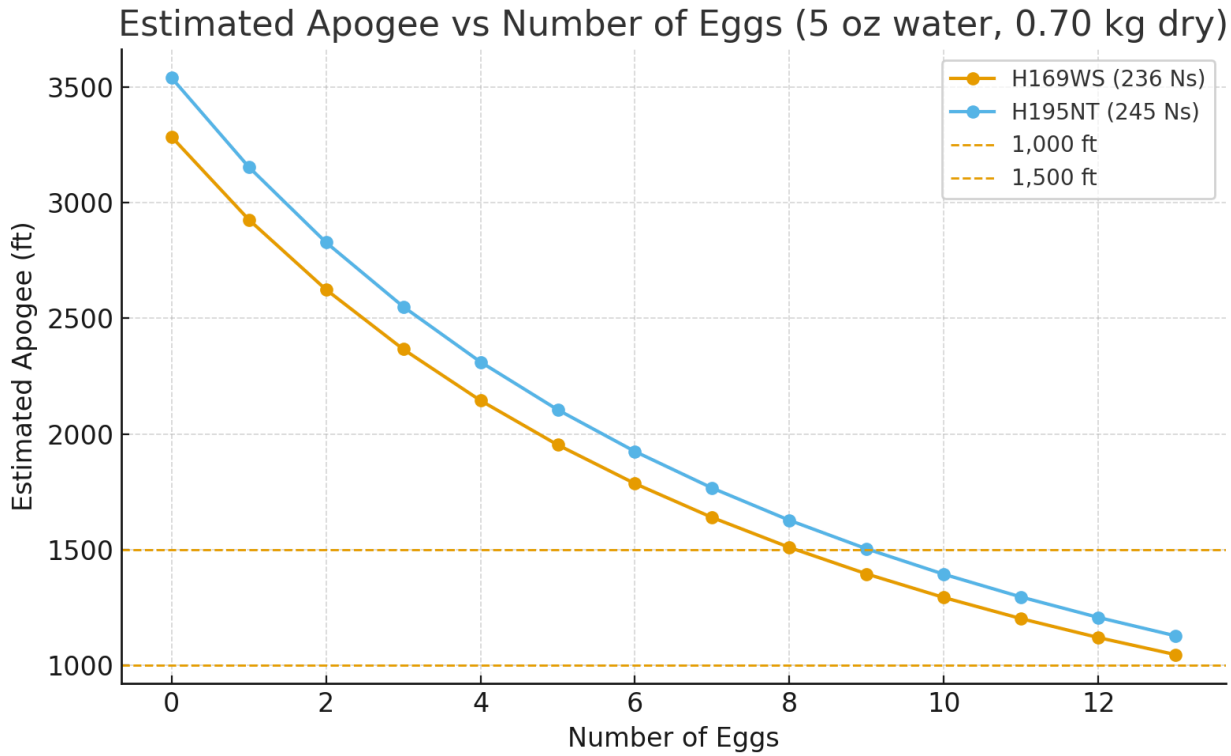


Fig 3. Estimated Apogee vs, Egg Count when using two H class motors

Recovery

Safe recovery of model rockets is very important, as a projectile shaped object falling out of the sky at a high speed is dangerous for the people below. To ensure safe recovery and a slow

descent, most model rockets employ the use of a parachute to slow them down enough to mitigate damage to the rocket or nearby objects.

In our case, we need a parachute that can slow the rocket down enough so that when it impacts the ground, both the rocket and payload are undamaged. As defined in the project requirements, a decent rate exceeding 25 feet per second during the final 5 seconds of flight will not be scored, so our descent must be slower than that. However we also have the previously stated constraint that the rocket cannot travel more than 800 feet from the launch site. A large parachute, while ensuring the safest landing, may also result in the rocket drifting a greater distance from the launch site. Our goal is to find a balance between the slowest descent and the least possible amount of drift.

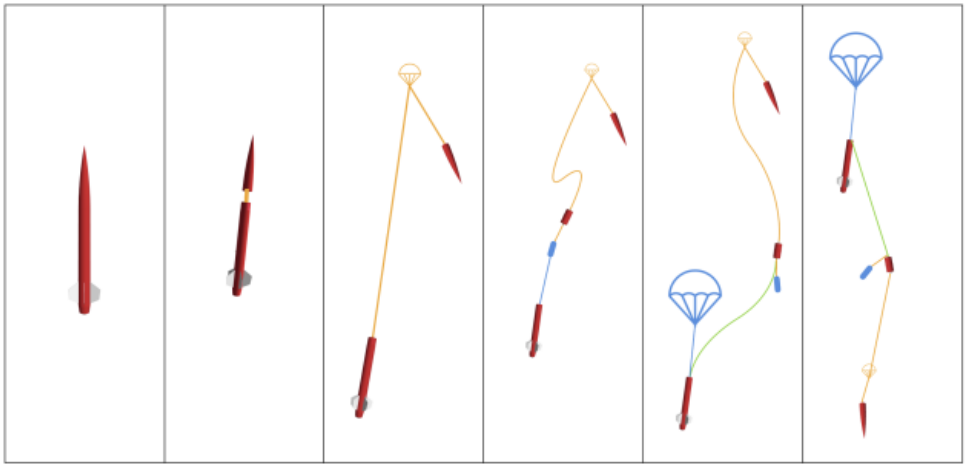


Fig 4. How the drogue and main parachutes are deployed in sequence

One of the best ways to create this balance is by having the parachute deploy at a lower altitude. The most common way this is done in model rocketry is with dual deployment. This method utilizes two separate parachutes, a drogue and a main, to slow the rocket's descent. A drogue parachute is a small parachute meant for deployment at high speeds, that can reduce the falling speed of the rocket, without letting it drift too far, but the drogue itself is not large enough to slow the rocket down to an appropriate and safe landing speed. Once the rocket reaches a low

enough altitude, we can have it deploy the main parachute. This parachute will be significantly larger in order to slow the rocket down to a safe landing speed which will ideally lead to no damage to the rocket and payload. Late deployment of the main parachute will allow for the desired balance of slowing the rocket while not allowing it to drift beyond 800 feet. Figure 4 models the deployment process of both the drogue and the main parachutes.

Another important consideration about the parachute is the need for a shock cord. The shock cord is what connects the parachute to the rocket, and it must be strong enough to absorb the force of slowing the rockets down without snapping. When the parachute is ejected from the rocket, the shock cord will rapidly unravel as the parachute expands and increases drag. Once the shock cord is fully unraveled and the parachute is fully deployed, the falling velocity of the rocket will instantly decrease, and this force will go directly into the shock cord.

Shock cords used in model rocketry are most commonly made out of nylon or kevlar. The pros of nylon are that it is cheap, flexible, and has a high elasticity. The flexibility allows for the shock cord to be easily packed into the rocket without taking up much space, and the elasticity allows the shock cord to stretch in order to absorb the force from parachute deployment. However, nylon may not be strong enough to absorb the impact of slowing down a rocket with a heavy payload such as ours and may snap upon parachute deployment. Kevlar on the other hand is also flexible, has a very high heat resistance, and it is extremely strong. The heat resistance is beneficial because it will be near the motor which will get very hot, and the strength of kevlar will prevent it from snapping when the parachutes get deployed. However, this strength comes at an increased cost, which will have to be factored into the budget. Additionally, due to the high strength of kevlar, it is not uncommon that the cord can cut into the rocket body when the cord absorbs the shock of parachute deployment.

Due to all these factors, our team may find the best success with a 48” main parachute and a 12” drogue parachute, connected to the rocket via a kevlar shock cord.

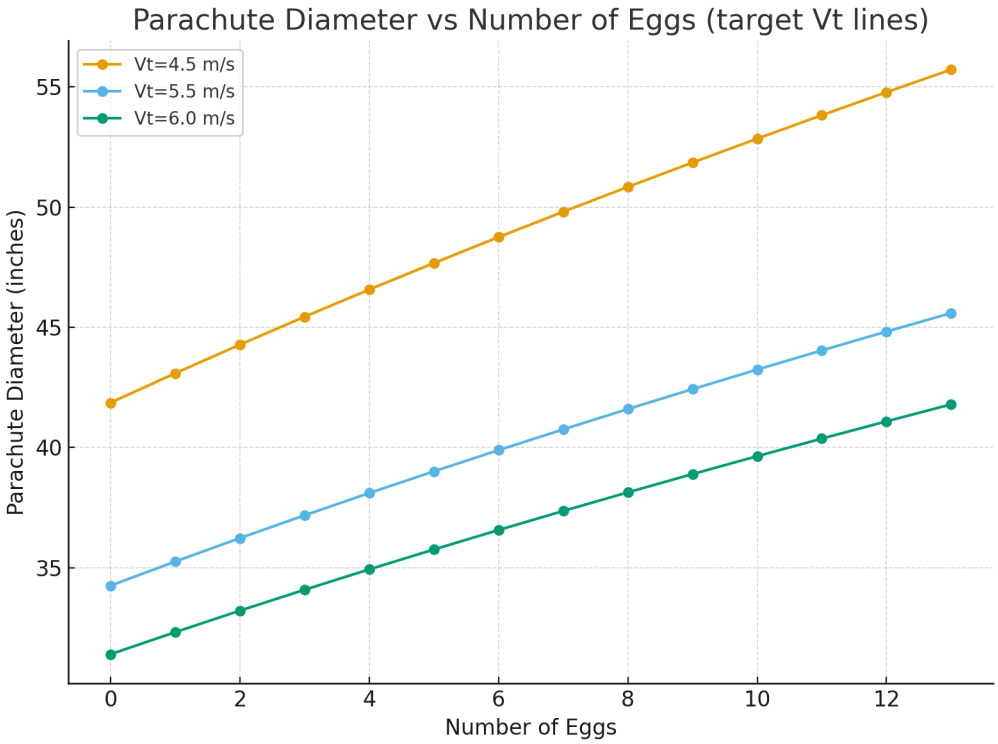


Fig 5. Parachute Diameter vs. Egg Count for different target velocities

Conclusion

Both engine selection and parachute selection are critical for a project based on rocketry. A rocket motor with adequate power to propel our rocket to the desired altitude, and a large enough parachute to slow the rocket’s descent back to the ground enough where the payload is undamaged are imperative to our team’s success in this challenge. To reach the desired altitude, we are likely to use an F, G, or H motor. To achieve the maximum possible points with the largest payload, we are likely to settle on an H motor. To initially slow the rocket’s descent without letting it drift too far, we will use a drogue parachute to assist in velocity reduction at higher altitudes, with the main parachute deploying at a lower altitude to slow the rocket to a safe

landing speed. We will also use a kevlar shock cord to attach the parachute to the rocket to ensure that our shock cord does not break on parachute deployment. With these methods of propulsion and recovery, we should be able to safely launch and land our rocket successfully and with optimally no damage to the payload.

References

[1] Tripoli Rocketry Association, “Tripoli Rocketry Association Safety Code”,

https://www.tripoli.org/safetycode

[2] Apogee Components, “Motor Comparison: Estes”,

https://www.apogeerockets.com/Rocket_Motors/Estes_Motors

[3] Apogee Components, “Motor Comparison: AeroTech”,

https://www.apogeerockets.com/Rocket_Motors/AeroTech_Motors

[4] Apogee Components, “Motor Comparison: Cesaroni”,

https://www.apogeerockets.com/Rocket_Motors/Cesaroni_Propellant_Kits

[5] Rocket Motor Components, “AeroTech Master Motor Matrix”,

https://s3.eu-west-1.amazonaws.com/static.fw1.biz/Templates/170652/myimages/aerotech%20master%20motor%20matrix%2012-19-24_1734643280505.pdf

[6] Wikipedia, “Motor impulse by class”,

https://en.wikipedia.org/wiki/Model_rocket_motor_classification

[7] Apogee Components, “Peak of Flight: What Size Parachute Should I Use”,

https://www.apogeerockets.com/education/downloads/Newsletter496.pdf

Figures

Fig 1 - Apogee Components, “29mm Motors (Loadable)”,

https://www.apogeerockets.com/Rocket_Motors/AeroTech_Motors/29mm_Motors_Loadable

Fig 2 - AeroTech Rocketry, “AeroTech F26-6FJ”,

https://aerotech-rocketry.com/products/product_02c36d5f-2a25-18b1-34e4-2e84a74e3c85

Fig 4 - The Rocketry Forum, jhill9693, “Deployment plan:”,

https://www.rocketryforum.com/threads/deployment-bag-question-first-time-user.188860/